
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) architectures have become a central ingredient in scaling modern neural networks, yet how they differ mechanistically from their dense counterparts remains broadly underexplored. A substantial body of interpretability work has examined how dense models exploit *superposition* to represent more features than they have dimensions, and how superposition arises as a function of feature sparsity and feature importance. We argue that conditionally computed models such as MoEs resist explanation through this same lens. Across our experiments, neither feature sparsity nor feature importance drives the discontinuous phase changes seen in dense networks; instead, network sparsity (the ratio of active to total experts) emerges as the more general organizing principle for characterizing MoE representations. We introduce metrics for measuring superposition across experts that apply naturally to a wide range of expert configurations. Our findings indicate that models with greater network sparsity tend to exhibit greater *monosemanticity*. Building on this, we propose a general definition of expert specialization grounded in monosemantic feature representation rather than load balancing, and show that experts tend to organize around coherent feature combinations whenever initialization encourages it. Taken together, these results suggest that network sparsity may offer a broadly applicable route to more interpretable models without sacrificing performance, challenging the common assumption that interpretability and capability are fundamentally at odds.

1 INTRODUCTION

Mixture of Experts (MoEs) have become pervasive across state-of-the-art neural models, spanning systems as diverse as Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), valued primarily for the computational efficiency and performance gains they deliver at scale (Shazeer et al., 2017; Fedus et al., 2022). A broad line of subsequent work has refined routing (e.g., Expert-Choice routing) as well as training stability and transfer (e.g., ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Both theoretical and empirical results further indicate that learnable routers can discover latent cluster structure in data, offering general insight into why experts specialize (Dikkala et al., 2023). Yet despite this widespread adoption across architectures and application domains, MoEs remain poorly understood from a mechanistic interpretability perspective.

A range of interpretability-oriented approaches has sought to make expert behavior more transparent. Yang et al. (2025b) proposes MOE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity. Park et al. (2025) introduce MONET, scaling the number of experts to enable capability editing via expert activation. Yet these efforts largely target architectural changes; we still lack a general mechanistic account of how MoEs represent features, how experts shape superposition, and whether experts specialize on their own without extra regularization. Mu & Lin (2025) survey the broader MoE literature and single out mechanistic interpretability as a key open challenge.

A fundamental and recurring challenge in interpreting neural networks of all kinds is the phenomenon of superposition: the regime in which models represent more features than

they have dimensions. Superposition lets networks pack many sparse features into far fewer neurons, but at the cost of rendering individual neurons polysemantic and difficult to interpret.

MoE architectures introduce an additional and largely orthogonal axis to this problem: network sparsity. Unlike dense models, which activate all neurons regardless of input, MoEs of any scale activate only a fraction of their total parameters (Shazeer et al., 2017). Whereas dense models exploit feature sparsity by packing many sparse features into shared neurons, conditionally computed models can afford to be far more selective, potentially dedicating entire experts to specific feature combinations.

We investigate whether (1) MoEs generally exhibit less superposition than their dense counterparts, (2) there is a discrete phase change in the amount of superposition of a given feature in MoE experts as a function of its relative importance and overall feature sparsity, as observed in dense models, and (3) expert specialization can be understood more generally through the lens of feature representation rather than load balancing alone.

We approach these questions using deliberately simple models that extend Elhage et al. (2022)’s framework to the MoE setting, a choice that lets us isolate mechanisms likely to recur across richer architectures. Our key contributions are as follows: (1) unlike dense models, MoEs do not exhibit sharp phase changes, instead displaying more continuous transitions as network sparsity increases; (2) MoEs consistently exhibit greater monosemanticity (less superposition) than dense models with matched active and total parameters, with individual experts representing features more cleanly; and (3) we propose an interpretability-focused definition of expert specialization grounded in monosemantic feature representation, showing that experts tend to organize around coherent feature combinations rather than arbitrary load balancing.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis holds that networks broadly represent concepts as directions in activation space (Park et al., 2024), yet across models the number of interpretable features routinely exceeds the available dimensions. Elhage et al. (2022) formalized this phenomenon as superposition, showing that dense models rely on non-orthogonal feature packing to maximize capacity at the cost of polysemanticity. While superposition is theoretically efficient (Scherlis et al., 2025), it forces reliance on complex post-hoc disentanglement methods, such as Sparse Autoencoders, to recover monosemantic features (Bricken et al., 2023). A growing body of work has examined how data correlations shape superposition (Prieto et al., 2025), and how interference patterns emerge and can be mitigated (Gurnee et al., 2023). We extend this broader line of inquiry to conditionally computed MoE architectures, showing that network sparsity—rather than feature sparsity alone—governs representational strategies.

Interpretability of MoEs. While MoEs have become a standard tool for scaling large models across many settings (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations continues to lag behind that of their dense counterparts. Existing analysis largely targets macroscopic behaviors, such as routing stability (Zoph et al., 2022), expert choice statistics (Zhou et al., 2022), or latent cluster discovery (Dikkala et al., 2023). More recent interpretability-focused approaches attempt to force specialization through architectural constraints, such as predefined concept routing (Yang et al., 2025b) or scaling expert counts to match vocabulary sizes (Park et al., 2025). These approaches, however, tend to prioritize capability editing over explaining the intrinsic feature geometry that any MoE must learn. We address this gap by analyzing how experts shape superposition in general, showing that MoEs exhibit greater monosemanticity than dense models and proposing a feature-based definition of expert specialization that is not tied to a particular routing recipe.

3 BACKGROUND

A primary goal of Mechanistic Interpretability is to reverse engineer neural networks; a common strategy is to decompose model representations into a set of human-interpretable concepts termed ‘features’. Across a wide range of models, these features are often assumed to be linear, that is, any hidden state h can be described as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

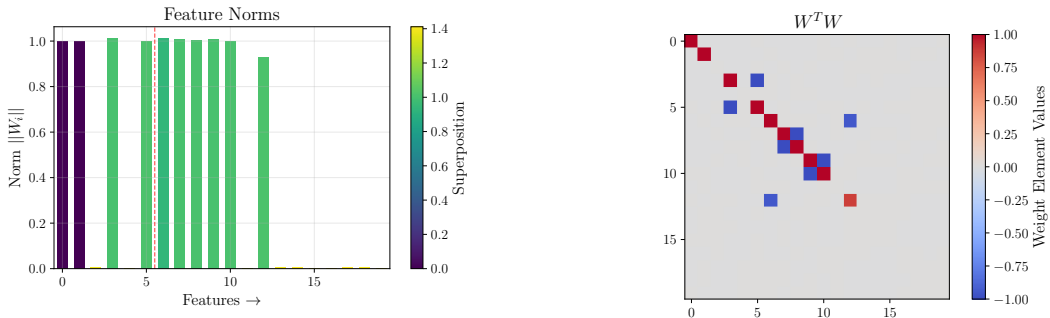
where $\vec{f}_i$ is the direction corresponding to feature i , α_i is the activation strength of this feature (roughly, the degree to which feature i is present in the input), and F is the set of all represented features.

The *superposition hypothesis* contends that models are capable of representing far more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). To fit many more features than dimensions into a latent space, the feature vectors in F must be packed so that they are not all orthogonal. When two features have interference $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say they are in *superposition*. Such superposition is tolerable so long as features are sparsely active—a property that holds broadly across natural data, from online text to many other modalities—though it comes at the cost of interpretability, since α_i then carries spurious activations unrelated to feature i actually being present in the input.

Monosemanticity is a property of individual neurons, in which a neuron’s activation corresponds cleanly to a single α_i (i.e., features are basis-aligned). When features are orthogonal (not in superposition) but not basis-aligned, neurons remain *polysemantic* even though feature interference is minimal. Throughout this work we concentrate on reducing superposition rather than enforcing basis-alignment. For brevity, whenever we describe models, experts, or features as “more monosemantic,” we mean they display less superposition.

4 DEMONSTRATING SUPERPOSITION

MoEs are frequently conceptualized as compositions of dense models, with each expert treated as an independent dense network. Whether experts actually represent features in the same way dense models do, however, remains an open and general question. We investigate it directly by comparing how MoEs and dense models differ in superposition.



(a) Norm of each feature’s weight vector $\|W_i\|$, with colors indicating superposition status (green for features in superposition, purple for monosemantic features).

(b) $W^T W$ matrix where each cell represents $(\hat{W}_i \cdot W_j)$, revealing interference patterns between features.

Figure 1: Feature representation and superposition in a representative dense model with $n = 20$ features and $m = 6$ hidden dimensions, with importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. Superposition (color) is given by $\sum_j (\hat{W}_i \cdot W_j)^2$.

4.1 EXPERIMENTAL SETUP

Our aim is to understand how a MoE can project a high-dimensional vector $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and then accurately recover it—a compression-and-reconstruction problem that recurs, in one form or another, throughout deep networks. This setup extends the framework of Elhage et al. (2022) to the MoE setting.

The input distribution. The input vector x stands in for the activations of an idealized, disentangled model, in which each dimension x_i corresponds to a distinct, independent feature—effectively a perfectly neuron-aligned, monosemantic representation of the kind interpretability aims to recover. We take $x_i \sim U(0,1)$, except that for a given sparsity $S \in [0,1)$ we set $P(x_i = 0) = S$. Because features rarely contribute equally to the loss in practice, we assign each feature x_i a scalar importance I_i that weights the reconstruction loss. Additionally, to isolate and study phase transitions for a single feature, we scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, so that $I = (1, 1, \dots, r)$. Together, I_i and r vary the signal strength of the features, letting us probe expert sensitivity to feature magnitude across a range of regimes.

Model Architecture. The MoE comprises E experts, where each expert e is parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^m$. Inputs are assigned to the top- k experts via a learned router $g(x) = \text{softmax}(W^r x)$ with $W^r \in \mathbb{R}^{E \times n}$, mirroring the routing mechanism used throughout practical MoE systems. Each active expert projects the input to a lower-dimensional hidden state $h^e = W^e x$ and generates a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The final output is the weighted sum of the active experts: $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$ where w_e are the renormalized gating weights.

We train our models with an L2 reconstruction loss weighted by the feature importances I_i , given by $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, to prevent expert collapse, we add the standard auxiliary load balancing loss (Fedus et al., 2022), defined as $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the total number of experts, f_e is the fraction of samples in a batch routed to expert e , P_e is the average gating probability assigned to expert e across the batch, and $\alpha = 0.01$ controls the penalty strength. We deliberately omit auxiliary load balancing in Section 4 so as to isolate the intrinsic architectural bias of the MoE toward superposition, independent of any balancing objective.

4.2 MEASURING FEATURE CAPACITY

To analyze feature representations across architectures, we compared two fundamental and architecture-agnostic properties of features: their *representation strength* and their *interference* with other features. We measured the norm of a feature weight vector in an expert e , given by $\|W_i^e\|$, which reflects the extent to which a feature is represented within expert e : $\|W_i^e\| \approx 1$ when feature i is fully represented in expert e and zero when it is not learned. We visualize the interference of a feature i with other features in expert e using the Gram matrix $W^\top W$, whose off-diagonal elements capture pairwise interference.

As shown in Figures 1a and 2a, the dense model and the MoE represent a comparable number of features (10 vs. 8) with similar norms for equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$). While the MoE experts show some local superposition (e.g., Expert 0 in Fig. 2b), the global interference structure is strictly partitioned. Unlike the dense model (Figure 1b), where any feature can interfere with any other, the MoE enforces a block-diagonal structure in which features routed to different experts share zero interference. This illustrates a general strategy: MoEs allocate representational capacity by partitioning the feature space, thereby narrowing the global scope of interference.

¹The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where $x_i \in [0,1]$ when nonzero. Furthermore, the off-diagonal terms in $(W^e)^\top W^e$ create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

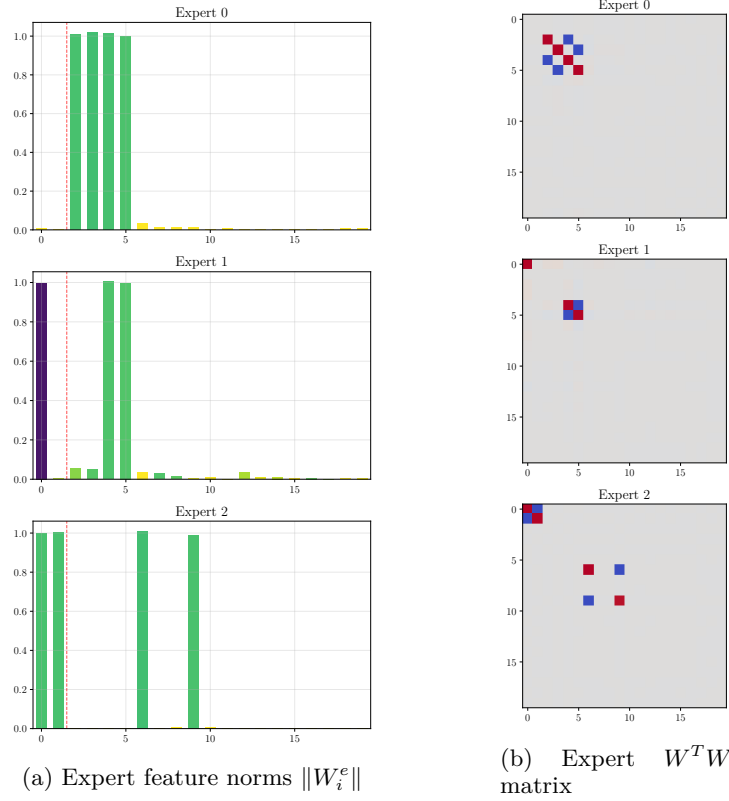


Figure 2: Feature representation and superposition in a representative MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert (top- $k = 1$ routing), with importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

Expert Feature Dimensionality. We next ask how MoEs allocate their limited representational capacity differently from the dense model. To this end we measured feature dimensionality, which quantifies the “fraction of a dimension” that a specific feature receives in a model (Elhage et al., 2022). For a feature i , we define its dimensionality in expert e by

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e is bounded between zero (not learned) and one (monosemantic). The total capacity of a MoE can thus be written as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. When features are “efficiently packed” in a model’s representation space, the dimensionalities of all features sum to the number of embedding dimensions, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For a MoE, the relation generalizes to $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, indicating that MoEs pack features just as efficiently as dense models given the same total parameters.

Features per Dimension & Network Sparsity. Since both dense and MoE models “efficiently pack” features in their representation space, we compared how the number of features per dimension differs across models. This lets us measure superposition precisely in both settings, and characterize how the number of experts in a MoE governs superposition across a broad range of feature sparsities. Whenever features per dimension exceeds one, the model is in superposition, since it represents more features than it has dimensions. We define features per dimension for a MoE by

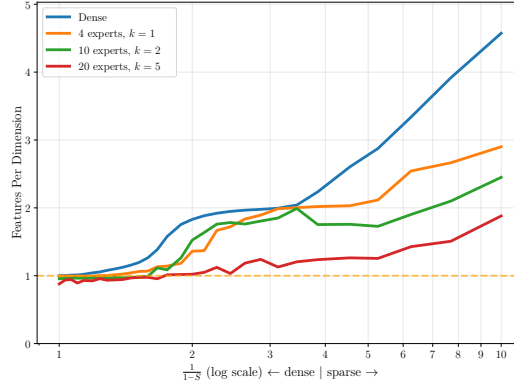


Figure 3: Features per dimension versus inverse feature density ($\frac{1}{1-g}$) for dense and MoE architectures under uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the most superposition, and superposition falls steadily as the expert count grows: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); 20 experts with $m = 1$, $k = 5$ (red). All models share equal total parameters and similar k/E . The dashed line at 1.0 marks monosemantic representation.

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the Frobenius norm and p_e is the expected probability that expert e is used across a batch of input samples, i.e. the average renormalized gating weight after top- k routing.

For MoEs with the *same* number of total parameters as the dense model, we consistently observe that the dense model has more features per dimension (Figure 3), i.e. more superposition. This indicates that the dense model leans on superposition to represent a greater total number of features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE tends to cap its representation at the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). Moreover, as we raise the total number of experts—holding total parameters and the ratio k/E roughly fixed—features per dimension steadily declines, i.e. superposition drops. *The greater the number of experts, the less superposition in the model.* In other words, features become more monosemantic as the expert count grows, a trend that holds robustly across the sparsity levels we test. The dense model’s greater superposition does let it attain consistently lower reconstruction loss than the MoEs (Figure 6, Appendix A.1), with a gap at any given sparsity of roughly 0.03–0.08; but as experts are added, the MoEs close this gap and reach comparable loss. See Appendix A.2 for theoretical intuition.

5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute them across experts with less interference. This suggests, more generally, that network sparsity reshapes how features are allocated rather than how many are learned. We therefore examined how properties of the input distribution—feature sparsity and importance in particular—drive this allocation, and whether they induce the physics-inspired *phase changes* seen in dense models.

Models have only a finite set of ways to represent any feature: it may be ignored, superimposed, or represented monosemantically. A phase change is the observation that, in some regimes, sharp discrete boundaries separate these regions as functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). By varying the sparsity and relative importance of features in the input distribution, we can elicit qualitatively different behavior—for instance, greater feature sparsity encourages greater superposition. Analyzing the per-expert phase diagram of MoEs reveals that they, in general, employ different representational strategies from dense models.

Setup. We follow the same setup as Section 4.1, adding a load balancing loss to avoid specialization collapse—the failure mode in which experts and routers settle into local minima where certain experts are entirely ignored—in the setups where it arises. We study three configurations, all with one active expert ($k = 1$): (A) $n = 2, m = 1$; (B) $n = 3, m = 1$; and (C) $n = 3, m = 2$. For each, we report the per-expert phase diagram across feature sparsity and last-feature relative importance while sweeping network sparsity, increasing the expert count (E) up to the number of input feature dimensions (n).

Throughout this section we hold active parameters fixed rather than total parameters, so that $m_{dense} = km_{moe}$ (ignoring router parameters). We do so to enable a controlled comparison within model architectures; otherwise, any observed differences could be attributed to architectural changes rather than to the number of active parameters. As it happens, 4.C.1/1 has the same number of active parameters as 4.B.X/2. The latter, however, has only one hidden dimension ($m = 1$) with which to encode the same number of input features ($n = 3$) as the former’s two ($m = 2$), making it harder to use superposition as a lens on specialization.

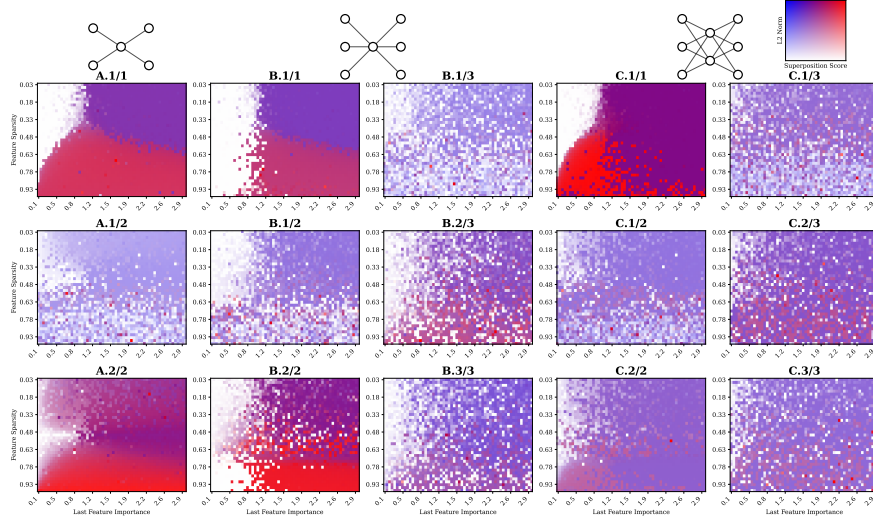


Figure 4: For a given expert and input dimension (feature), we can decode how that feature is embedded in the hidden dimension—whether it is ignored (white), monosemantic (blue-purple), or superimposed (red). We plot the joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j < n} (\hat{W}_n \cdot W_j)^2$) across a broad sweep of feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last of n total features. For each cell, we train ten models and keep the one with the lowest loss. A load balancing loss is used throughout this section. We report the joint feature norm and superposition for the last feature: a low L2 norm ($\|W_n\|$) is white, denoting that the model ignores the last feature; otherwise a low superposition score is blue-purple, indicating a monosemantic representation of the last feature, while red indicates the feature is represented in superposition. Cell (i, j) in subfigure X.e/E denotes expert e of E total experts trained on architecture X for last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 indicates a dense model.

Results and Takeaways. In every single-expert (dense) case, we observed a clear phase change (Figure 4.X.1/1), affirming the findings of Elhage et al. (2022). As soon as we increased the total number of experts, however, discrete phase changes disappeared. Some experts in MoEs with $E = 2$ resemble their respective dense cases (Figure 4.X.2/2) but

exhibit more continuous transitions. In each case, one expert grew more monosemantic, specializing in the feature of highest relative importance. Experts that diverge from the dense cases uniformly show much lower superposition scores (they are bluer), indicating more monosemantic representations. This is consistent with the previous section and points to a general tendency—MoEs favor lower superposition scores than their dense counterparts.

For the $n = 2, m = 1$ setup (Figure 4.A), the dense model fails to represent the last feature when feature sparsity is low, whereas the comparable MoE preserves it far more readily because it has the capacity to do so. With three input dimensions (Figure 4.B), MoEs no longer show this behavior, since the experts are already superimposing the other two features and no room remains for the third within a single hidden dimension. Unlike the other two cases, for architecture B a single hidden dimension in superposition is insufficient, in the high-sparsity regime, to represent all features. Even so, we see no clear phase change—except in the 0.5–0.7 feature-sparsity region of 4.B.2/2, where the pattern is mostly discrete but mixed. For $m = 2$, the white region in the dense model (Figure 4.C.1/1)—in the mid- to low-feature-sparsity domain, where the feature is relatively less important than the others—reflects the last feature being ignored. As network sparsity increases, however, across the remaining panels of Figure 4.C the models represent the last feature with progressively greater L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Put differently, dimensionality in the low relative-importance region rises with network sparsity, echoing the general trend of Figure 3.

Across Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 we observed a window of feature sparsity from roughly 0.48 to 0.7 in which polysemanticity, monosemanticity, and outright ignorance are heavily intermixed. This points to a middle ground in MoEs where polysemantic and monosemantic representations achieve comparable loss, making it hard for the model to commit consistently to the strategies seen in the low- and high-sparsity domains. We observe no analogous pattern in dense models—further evidence that MoEs, as a class, learn representational strategies distinct from those of dense networks.

Conclusion. These experiments all use $\text{top-}k = 1$, so only one expert is active at a time. Even under this most dense-like regime, we observe markedly different behavior already at $E = 2$ —including when a single hidden dimension in superposition would suffice to represent all features. We therefore conclude that it is misleading, in general, to think of MoEs as mere aggregations of dense models. The router mechanism, by exposing each expert to only a subset of the feature domain, fundamentally reshapes how the experts behave and learn.

6 EXPERT SPECIALIZATION

Given that MoEs exhibit less superposition, we now turn to how these monosemantic features are organized within experts and what that organization implies for specialization.

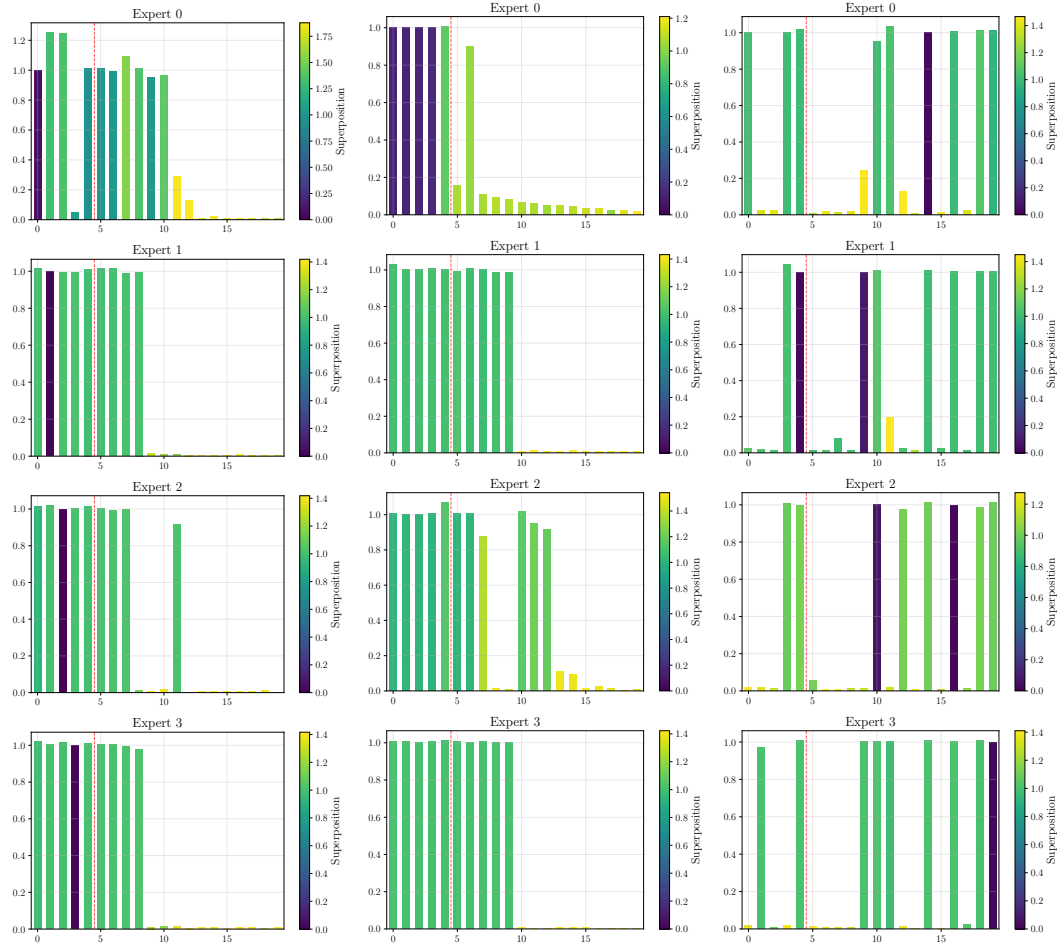
Expert specialization in MoEs is traditionally framed around load balancing across all inputs (Chaudhari et al., 2025). Yet this framing misses the more natural, general intuition of specialization: that an expert should be invoked precisely when the concepts it specializes in are present in the input.

We define an expert as specialized if it *occupies* certain feature directions in the input space and represents those features relatively monosemantically. We show that these two conditions are directly correlated, and that their joint presence naturally encourages load balancing across experts—recovering the conventional notion of balance as a consequence rather than an imposed objective.

Because we fix $k = 1$, the feature space is partitioned into convex cone regions (see Appendix A.3), each routed to a particular expert. By definition this means $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points contained within the cone and s is any positive scalar. If a feature vector $\vec{x}$ is routed to an expert, then every $s\vec{x}$ is routed to the same expert. In this case we say that feature x is contained within expert e , and accordingly that expert e *occupies* x .

We empirically find that small models which distribute the input space across more experts tend to achieve lower loss (see Appendix A.4). This raises a broader question: does the way the input space is allocated to experts imply anything about those experts’ representations? Our definition of expert specialization predicts that this allocation implies *monosemanticity*—a correlation that, as we will see, continues to hold as we scale up to larger toy models (e.g., $m = 10$).

In models with $n > 2$, we explore two complementary directions: first, whether initializing experts to occupy features in the input space causes those experts to become more monosemantic with respect to those features; and second, conversely, whether—for the features an expert chooses to represent monosemantically—that expert also comes to occupy those features in the input space.



(a) Diagonal Initialization (b) Ordered K-hot initialization (c) Random K-hot initialization

Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) for three different initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^T = \hat{e}_i$, the basis vector for that dimension), with relative feature importance decreasing exponentially in order from feature one to 20. In (b), the gate matrix uses an “ordered k-hot” initialization, so that the first expert aligns with the first five features and each subsequent expert aligns with the next five; relative feature importance is as in (a). In (c), the gate matrix uses a “random k-hot” initialization, in which each expert is assigned five random features such that experts share no common feature yet collectively cover all 20; relative feature importance again decreases exponentially but is distributed randomly across features.

When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied, and only that feature (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the other experts, initialized over other features, represented neither those less important features nor the five most important features they were not initialized over. When we break the ordering of feature importance and randomize the features each expert initially occupies, each expert monosemantically represents only the most important feature(s) it was initialized over (Figure 5c). These consistent outcomes across all three initialization schemes suggest the underlying mechanism is general rather than an artifact of any single scheme.

There is a strong and consistent correlation between the features initially routed to an expert and the features that expert ends up representing monosemantically. We further observe that experts monosemantically represent only important features. This holds whether we explicitly initialize each expert with a single important feature, or hand it a set of features from which it selects the most important one itself and represents it monosemantically.

Even under uniform feature importance, where experts place all features in superposition with higher levels of interference, the pattern persists: the features an expert initially occupies remain *relatively* more monosemantic, on average achieving superposition scores 1.03 standard deviations below that expert’s mean.

We next investigated whether experts that represent certain features monosemantically also occupy those features in the input. To do so, we measure usage statistics when those features are *one of many* active features, and when they are the *only* active features; this second case is equivalent to measuring the probability that the expert occupies those features. The correlation holds under both xavier and k-hot initialization schemes (Table 1), suggesting it does not hinge on a particular initialization. Given $E = 10$, a mean expert usage of $\sim 10\%$ indicates even load balancing across experts. In every case, whenever an expert’s corresponding monosemantic feature(s) is active, that expert’s usage rises substantially; when the feature(s) is the sole active feature, the expert dominates usage. Under the k-hot initialization scheme, 100% of all features monosemantically represented by an expert are occupied by that same expert.

As experts represent more features monosemantically, they can be regarded as more specialized. Their usage on arbitrary input decreases, yet conditional on their specialized features being active their usage rises far more sharply than that of other experts. This pattern holds across all cases, the lone exception being the xavier-initialized model’s four-monosemantic-feature expert, which shows a pronounced drop in utilization.

7 CONCLUSION

We investigated how experts shape superposition in MoEs, showing that MoEs consistently exhibit greater monosemanticity than dense networks while forgoing the phase changes characteristic of dense models. We proposed a feature-based definition of expert specialization and demonstrated that experts naturally organize around coherent features whenever initialization encourages such specialization. These findings are established in simple autoencoder toy models with synthetic data, which lets us isolate the mechanisms cleanly while leaving open how far they extend to large-scale transformers with unknown feature distributions (see Appendix A.6). Even within this controlled setting, our toy MoEs achieve comparable loss while maintaining markedly more interpretable representations—challenging the prevailing view that mechanistic interpretability and model capability are fundamentally in tension. We view this as evidence that network sparsity may be a broadly useful lever for interpretability. Promising directions for future work include identifying what favors monosemanticity in MoEs, how the training dynamics of MoEs diverge from those of dense models, and when specialization emerges. Answering these questions could inform the design of more interpretable, high-performing models across the many domains where MoEs are now deployed.

Table 1: Monosemantic-feature and usage statistics per expert for $n = 100, m = 10, E = 10$. We train one hundred models for each initialization scheme (xavier and k-hot), yielding 1000 experts apiece. Each statistic is aggregated across models, with experts grouped by the number of features they represent monosemantically. For the feature(s) an expert represents monosemantically, we track expert usage both when that feature(s) is one of several active features in the input and when it is the *only* active feature.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

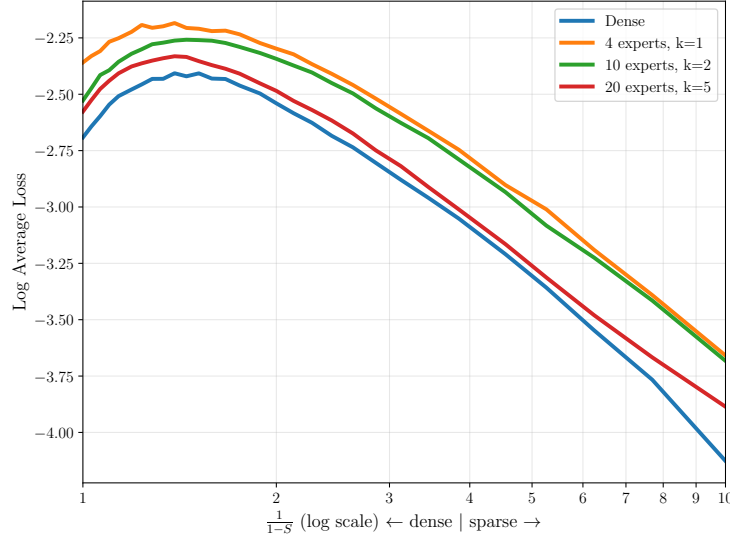


Figure 6: Log average loss versus feature density ($\frac{1}{1-S}$) for dense ($m = 20$) and MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all with uniform feature importance ($I_i = 1.0$) for $n = 100$ input features. Results are averaged over five runs per sparsity level. Although the dense model outperforms every MoE at all sparsity levels, the MoE loss steadily approaches that of the dense model as the number of experts grows.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs, by contrast, “eat this same gap” structurally through conditional computation. By aligning activation sparsity with feature sparsity, MoEs remove the computational penalty for maintaining dedicated, rarely-activating neurons. As Elhage et al. (2022) note, once a model expends computation only on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy—in effect trading the *compression* of superposition for the *selection* of routing. We expect this argument to apply well beyond our toy setting, to any architecture that conditions computation on the input.

Geometrically, this structural change manifests as a reduction in *interference*. While the global ratio of represented features to parameters stays constant, the router partitions the feature space so that a feature routed to expert e competes for capacity only with the subset of features also assigned to that expert. Interference is therefore governed by the local expert matrix $(W^e)^\top W^e$ rather than the global $W^\top W$ of a dense model. This partitioning sharply reduces the number of interfering features for any given feature vector, easing the optimization pressure to pack features in superposition and allowing experts to learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the $k = 1$ regime, the router function $g(x) = \text{softmax}(W^r x)$ reduces to $\text{argmax}(W^r x)$. The region routed to expert i is then $\forall j \neq i, (w_i - w_j)^\top x > 0$, where w_i and w_j are row vectors of W^r . Each such constraint is a homogeneous linear inequality, and regions bounded by them are by definition convex cones: if a given x satisfies the inequality, then scaling

both sides by any positive scalar s preserves it, and if x_1 and x_2 both satisfy it, so does any convex combination $x = x_1\lambda + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$.

For $k > 1$, the region of inputs routed to a particular expert e becomes a union of convex cones. Since the union of convex cones is not in general itself a convex cone, our characterization of experts as occupying feature directions may not extend beyond $k = 1$ —a boundary we flag rather than assume away.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

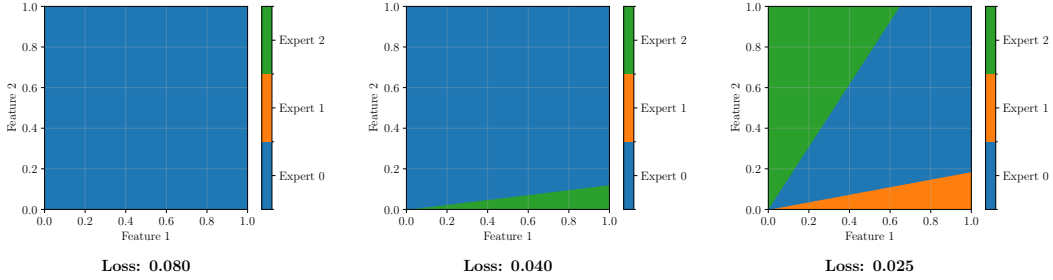


Figure 7: Expert routing of three identical models under differing initialization schemes, with $n = 2$, $E = 3$, $m = 1$. The first model (left) performs worst (loss: 0.08), collapsing all inputs onto a single expert. The second (middle) improves (loss: 0.04), routing a small portion of inputs—those in which feature 1 is active—to a second expert. The third (right) achieves the lowest loss (0.025) by distributing the input space across all experts: one expert is chosen when only feature 1 is active, one when only feature 2 is active, and one when both are. The trend illustrates the broader relationship between input-space distribution and performance.

In small models ($n = 2$, $m = 1$, $E > 1$), we consistently find—across $E \in [2, 7]$ —that models distributing the input space over more experts tend to achieve lower loss. Fixing $n = 2$ lets us directly visualize which portions of the input space are routed to which experts, as shown in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setting of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical map from any dense model to a monosemantic MoE with an equivalent number of active features under a sparsity constraint. We present it here as a general construction whose assumptions make explicit the conditions under which it holds.

Assume an upper bound a on the number of active features for any input, so that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Assume further that a is no greater than the hidden dimensionality m of an expert, bounding the number of features a model must represent, and that the hidden dimension is smaller than the total number of input features n ($a \leq m \leq n$). To construct the monosemantic MoE, for each possible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —i.e., every set of active features whose size is at most a —create an expert that monosemantically preserves those features. (In fact, it suffices to take only the subsets with $|S| = a$.) The router then selects the expert corresponding to the active features, of which there are never more than a by assumption:

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert has ample capacity to represent its assigned features without superposition. To restate the construction: at most a features are ever active, and every distinct combination of active features receives its own dedicated expert with enough capacity to represent those features monosemantically. The number of experts required is thus $\binom{n}{a}$.

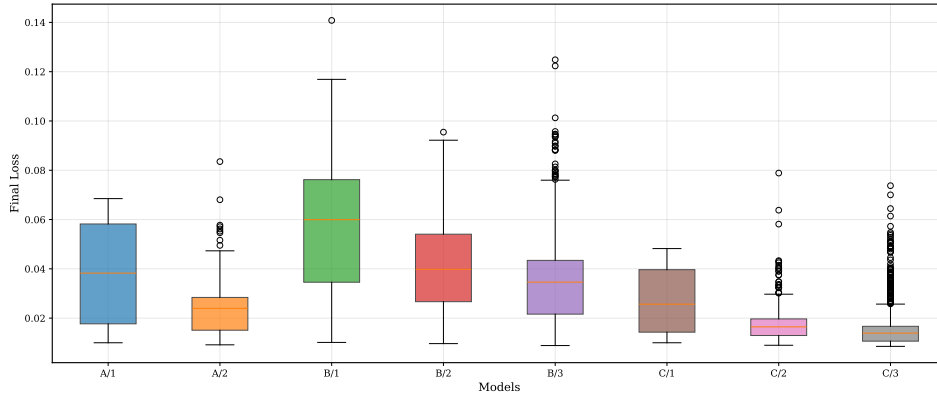


Figure 8: In model X/E , X denotes the same architectures and models used in Figure 4 and E denotes the total number of experts (i.e. network sparsity). Increasing network sparsity lowers mean loss while raising localized variance—an effect that grows as the number of experts approaches the input feature dimensions. We attribute the added variance to the relatively unstable training of MoEs compared to dense models, which persists even though we train ten models per cell and keep the lowest-loss one.

This theoretical MoE attains zero reconstruction loss only while the sparsity constraint holds (or, in these toy models, as sparsity goes to one), because more than m features could occasionally be active at once ($a \not\leq m$), exceeding the network’s monosemantic capacity. Crucially, though, the dense polysemantic model fares no better in this case unless features are correlated in the distribution. Hence, even when $a \not\leq m$ on some inputs, the polysemantic model faces the same obstacle, and the monosemantic MoE under this construction may still outperform it under looser sparsity constraints.

In short, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ under the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ with $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs; in the toy settings of this paper, the constraint holds in the limit where sparsity goes to one. More broadly, in practice there may be an upper bound on how many features a given quantity of information can semantically encode, reflected in the size of meaningful embeddings of that data. A MoE with enough experts and a tractable—hence interpretable—amount of superposition may therefore suffice to encode all features present in real data.

A.6 LIMITATIONS

Our results should be read in light of several limitations that also delimit the scope of our claims. First, all experiments are conducted in controlled toy-model settings adapted from superposition studies in sparse autoencoders (Elhage et al., 2022). This design enables precise measurement of interference and monosemanticity, but it abstracts away many complexities of large-scale transformers—including multi-step computation, attention, heterogeneous feature distributions, and realistic routing dynamics—so the transferability of our findings to deployed MoE systems remains to be established.

Second, our architectural and routing choices are deliberately restricted: simple experts, fixed top- k routing (often $k = 1$), matched parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs frequently adopt richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or move beyond reconstruction to objectives such as next-token prediction, any of which may induce qualitatively different representational behaviors.

Third, the feature distributions in our synthetic tasks—their sparsity patterns, feature importance, and independence assumptions—are considerably simpler than those of natural data across most domains.

Finally, while we observe increased monosemanticity and reduced superposition in MoEs under fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale, all of which prior work suggests can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Overall, our study offers mechanistic insight under clean experimental conditions, and we regard it as a foundation to be validated in larger, more realistic MoE systems. Because our current methods do not scale naturally to such models, extending them is a clear and important direction for future work.